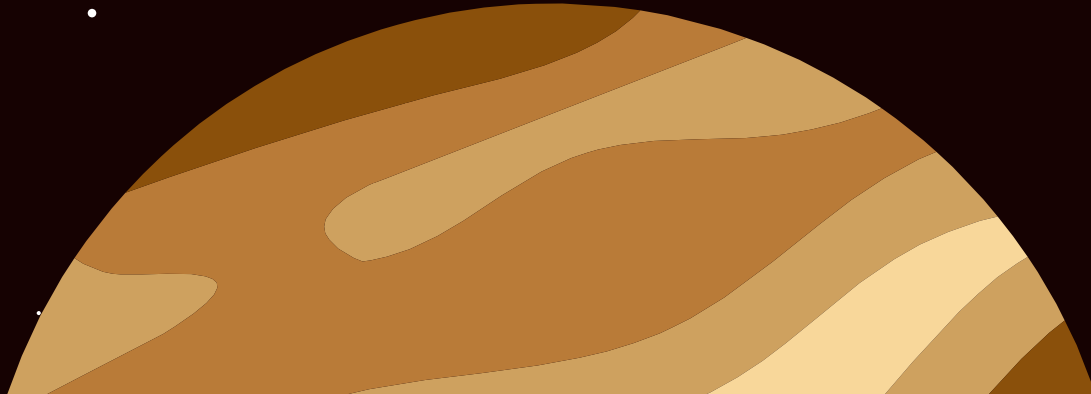


Martian Subsurface Analysis:

Utilizing Insight Seismic Data to Detect Liquid Water and Brine Deposits

Addy Peterson, Petra Elrod, Shelby Layne



INTRODUCTION

Insight Mission

- Formally: Interior exploration using seismic investigations, geodesy, and heat transport (Lockheed Martin, 2018)
- Recorded 1,313 Marsquakes (JPL, 2022)
 - Mars has no tectonic plates
 - Marsquakes formed by meteorite impacts and planet shrinking as it cools
- Seismic data from Marsquakes produces a picture of the Martian subsurface and provides evidence for water on Mars

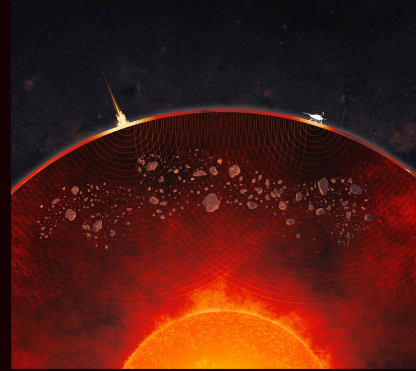
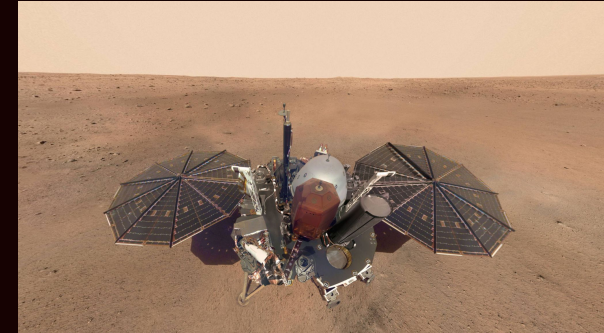


Figure 1: *Insight detecting seismic waves* by NASA (28 August 2025),

[<https://science.nasa.gov/photojournal/impactor-debris-scattered-within-mars-mantle-artists-concept/>]

Figure 2: *Insight Selfie* by The Planetary Society (11 December 2018),

[<https://www.planetary.org/space-images/insights-self-portrait-as-a>]



INTRODUCTION

Seismic Waves

- Seismic waves produce distinct signals through certain materials (PNW seismic Network, 2024)
 - P waves pass through liquids and solids
 - S waves only pass through solids
 - S waves require shear strength to propagate, which liquids lack
- Because of this distinction, we decided to analyze the velocity ratios of P and S waves in our project (V_p/V_s)

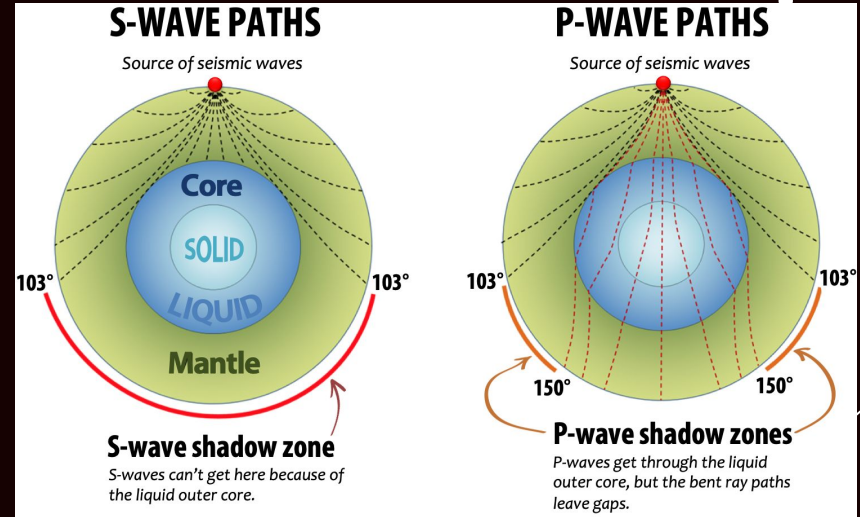


Figure 3: *Trajectory of Body Waves* by British Columbia Pressbooks,

[<https://pressbooks.bccampus.ca/physicalgeologyh5p/chapter/3-2-understanding-earth-through-seismology-2/>]

RESEARCH QUESTION AND HYPOTHESIS

Question

How can variations in V_p/V_s ratios from NASA's InSight data indicate volatile-rich zones in Mars' crust?

Hypothesis

Since S waves cannot travel through liquid, there should be a significant increase in V_p/V_s ratios when Marsquake seismic waves pass through liquid compared to when they pass through solid rock.

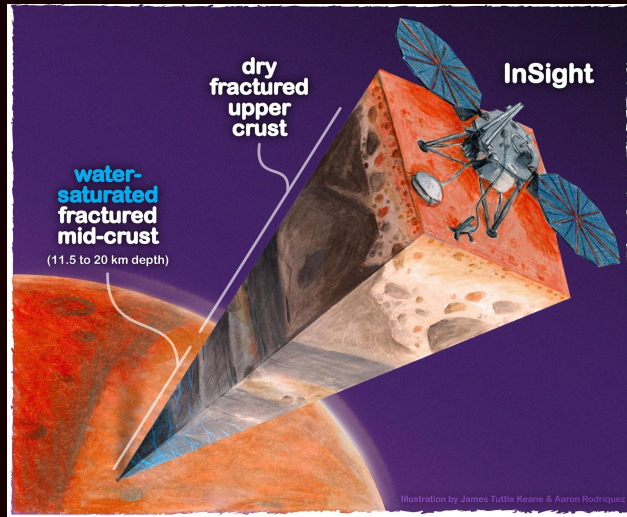


Figure 4: *InSight Analyzing Martian Subsurface* by James Tuttle Keane and Aaron Rodriguez,

[<https://scitechdaily.com/subsurface-oceans-on-mars-nasas-in-sight-uncovers-vast-reservoir-of-liquid-water/>]

OBTAINING THE DATA

Primary Source

NASA InSight Marsquake Service (MQS)

- PDS (Planetary Data System)

Formatting

QuakeML (XML) format:

- Events and IDs
- P / S Wave Arrival Times

Mars Quakes Catalog:

- Event ID
- Distance
- Quality
- Class

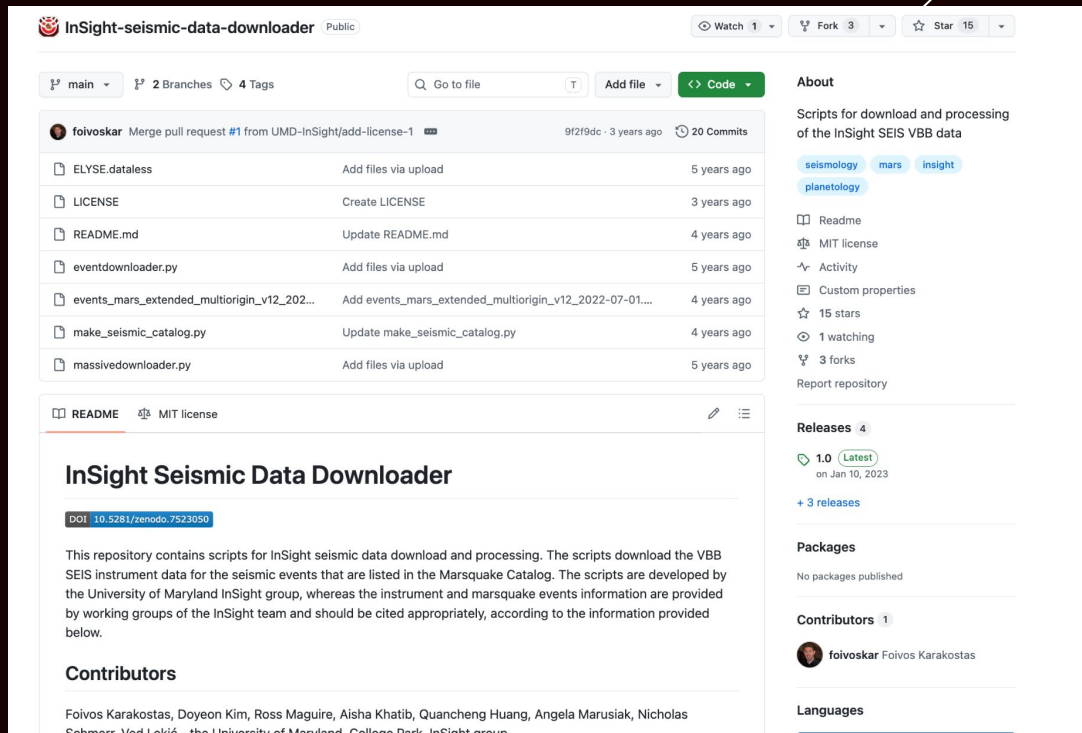


Figure 5: UMD - InSight Seismic Data Downloader Repository by Foivos Karakostas

[<https://github.com/UMD-InSight/InSight-seismic-data-downloader>]

PREPROCESSING



- Custom XML File Parser
- Python parser using the `xml.etree.ElementTree` library, to create a new Pandas Dataframe to work with:

- Extracted P and S wave arrival times
- Calculated Vp Vs Ratios
- filter to remove "unphysical" noise (ratios <1.4 or >2.4)
- Used A and B Quality Events
- Low Frequency and Broadband

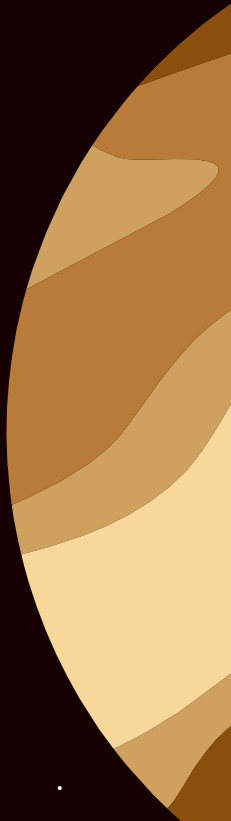
	Event_ID	Distance_Degrees	Distance_KM	t0	tp	ts	Vp_Vs_Ratio
5	S1022a	30.73	1819.216	2021-10-11T23:14:29.105382Z	2021-10-11T23:18:25.025302Z	2021-10-11T23:21:23.255908Z	1.755471
6	S1015f	27.49	1827.408	2021-10-04T04:52:29.248537Z	2021-10-04T04:56:00.185082Z	2021-10-04T04:58:40.218707Z	1.758681
7	S1000a	128.29	7594.768	2021-09-18T17:46:20.639751Z	2021-09-18T17:59:01.470476Z	2021-09-18T18:13:28.0Z	2.138926
8	S0976a	146.26	8658.592	2021-08-25T03:32:20.629953Z	2021-08-25T03:48:58.4Z	2021-08-25T04:03:07.778327Z	1.851277
10	S0820a	30.19	1787.248	2021-03-18T14:51:33.869889Z	2021-03-18T14:54:39.0Z	2021-03-18T14:57:39.0Z	1.972289
14	S1197a	31.98	1893.216	2022-04-09T22:05:52.41687Z	2022-04-09T22:09:38.5Z	2022-04-09T22:12:49.0Z	1.842610
15	S1192a	25.50	1509.600	2022-04-04T23:06:52.616746Z	2022-04-04T23:12:52.72401Z	2022-04-04T23:16:48.265836Z	1.861661
16	S1163a	23.10	1367.520	2022-03-06T02:29:13.668752Z	2022-03-06T02:34:18.697973Z	2022-03-06T02:38:28.753599Z	1.819776
18	S1157a	35.58	2106.336	2022-02-27T01:02:36.620313Z	2022-02-27T01:07:07.386079Z	2022-02-27T01:10:33.077426Z	1.759665
20	S1135c	20.00	1184.000	2022-02-05T04:34:04.201673Z	2022-02-05T04:39:24.217017Z	2022-02-05T04:43:00.872391Z	1.677016
21	S1106a	38.50	2279.200	2022-01-06T14:53:18.310024Z	2022-01-06T14:58:20.331235Z	2022-01-06T15:05:17.434669Z	2.381040
22	S1105a	22.40	1326.080	2022-01-05T07:59:04.008824Z	2022-01-05T08:04:13.988379Z	2022-01-05T08:08:16.344706Z	1.781846
23	S1094d	24.40	1444.480	2021-12-25T03:31:41.685086Z	2021-12-25T03:36:52.573319Z	2021-12-25T03:41:16.658691Z	1.849454
24	S1091a	20.20	1195.840	2021-12-21T23:57:27.099737Z	2021-12-22T00:02:46.930495Z	2021-12-22T00:06:25.465309Z	1.683283
27	S1086d	24.90	1474.080	2021-12-16T23:02:32.671075Z	2021-12-16T23:07:42.4055Z	2021-12-16T23:12:11.468547Z	1.868690
28	S1082c	21.20	1255.040	2021-12-12T20:20:50.278146Z	2021-12-12T20:25:58.383505Z	2021-12-12T20:29:47.920691Z	1.744996
29	S1077c	23.00	1361.600	2021-12-07T14:01:49.15588Z	2021-12-07T14:06:59.285347Z	2021-12-07T14:11:08.381859Z	1.803202
30	S1075c	25.60	1515.520	2021-12-05T14:43:47.693762Z	2021-12-05T14:49:10.902171Z	2021-12-05T14:53:48.242672Z	1.858086
31	S1073d	26.50	1568.800	2021-12-03T14:28:52.584603Z	2021-12-03T14:33:57.521266Z	2021-12-03T14:38:44.06911Z	1.939696
32	S1071b	25.00	1480.000	2021-12-01T12:19:05.491437Z	2021-12-01T12:24:25.3756Z	2021-12-01T12:28:55.650274Z	1.844914
33	S1068c	24.00	1420.800	2021-11-28T08:10:32.786857Z	2021-11-28T08:15:52.77573Z	2021-11-28T08:20:12.269144Z	1.810945
34	S1065b	22.50	1332.000	2021-11-25T11:06:55.004195Z	2021-11-25T11:12:04.938873Z	2021-11-25T11:16:08.182224Z	1.784821
35	S1060a	26.90	1592.480	2021-11-19T11:40:11.956993Z	2021-11-19T11:45:24.195699Z	2021-11-19T11:50:15.90031Z	1.934236
36	S1054c	26.20	1551.040	2021-11-14T03:24:25.479129Z	2021-11-14T03:29:31.120887Z	2021-11-14T03:34:14.706593Z	1.927837

Figure 6: processed seismic data frame by Petra Elrod

PARAMETERS

Data Selection Criteria

- **Defining the Volatile Zone (1.8-2.1)**
 - Studies of fluid-saturated porous rock
 - Silicate crust (~ 1.73) + liquid brines reduces shear wave velocity (V_s), down to 1.8–2.1 range
- **Poisson's Ratio**
- **Quality A & B Events**
 - Use events with clear, sharp phase arrivals to get solid ratio values without excess noise
- **Low Frequency (LF) and Broadband (BB)**
 - LF dives deeper into the crust (200km+), bypassing the noisy surface
 - BB events provide complete stable seismic record for our calculations



QUALITY ANALYSIS

Ratings: A, B, C and D

Seismic Phases

Visibility of P and S-wave arrival.

Direction

Can the direction be determined using computational methods?

- Polarization

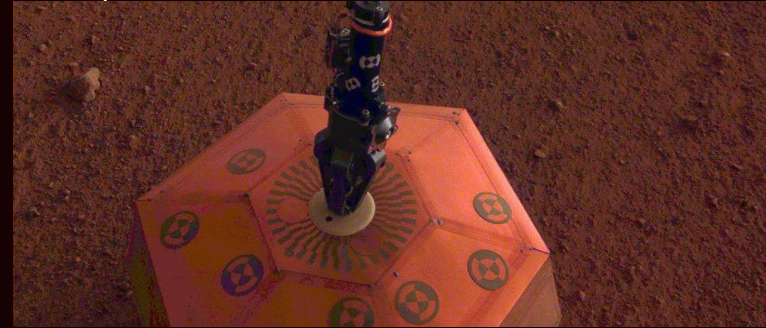


Figure [7]: NASA's InSight Lander by Universe Today (21 December 2018)
[<https://www.universetoday.com/articles/insight-just-placed-its-seismometer-onto-the-surface-of-mars-to-listen-for-mars-quakes>]

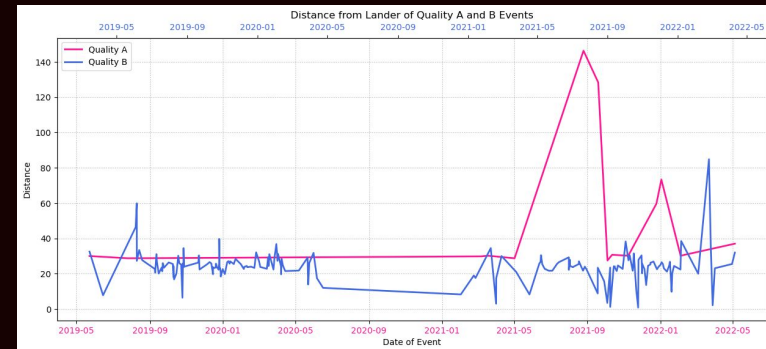


Figure [8]: Distance from Lander of Quality A and B Events by Addy Peterson (11 May 2026)

Subsurface Analysis

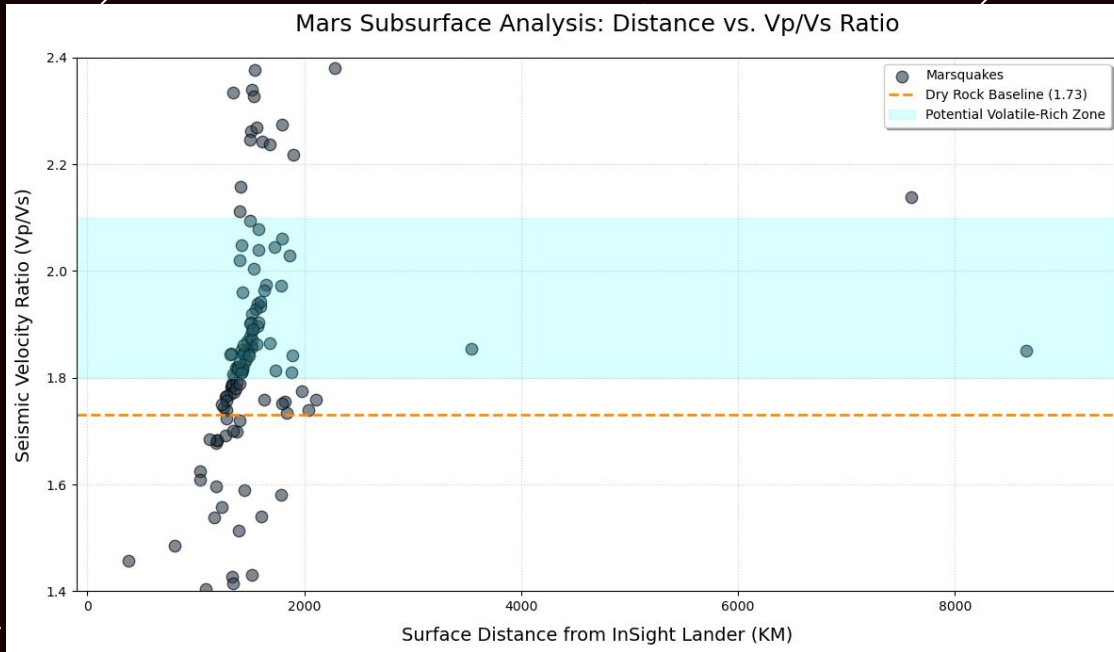


Figure [9]. Subsurface analysis Vp Vs Ratios vs Distance by Petra Elrod (1 May 2026)

- Relationship between Distance and Vp/Vs velocity ratio
- Vp Vs threshold of 1.8, cyan band (1.8-2.1) represents the volatile zone
- Dry Rock Baseline of 1.73 was determined with Poisson's ratio. On Earth and Mars, this is the standard baseline for dry, solid basaltic rock (the primary material of the Martian crust).

Subsurface Analysis

- We treated the InSight lander as the center of our map.
- Our high-ratio seismic events (the colored dots) here represented in degrees.
- This suggests that while the Martian surface is dry, a global 'mushy' layer still exists at depth.

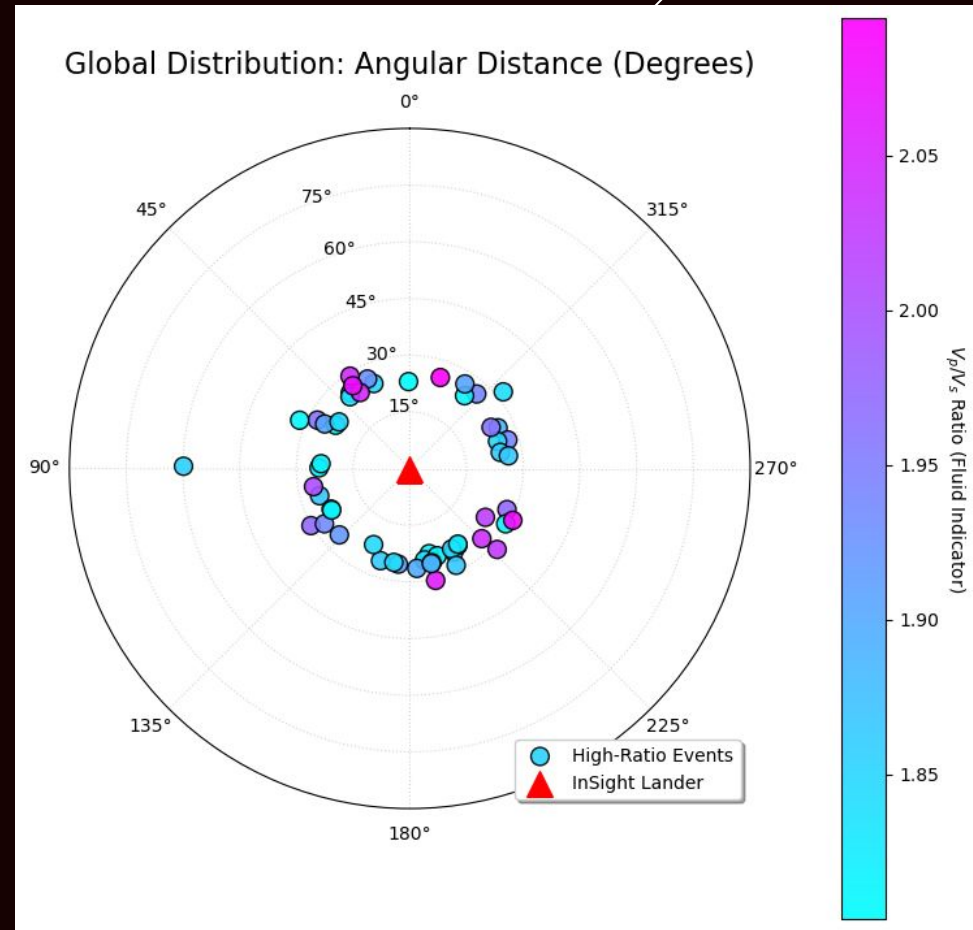


Figure [10]: Global Shell Model - Subsurface analysis in degrees from the lander
by Petra Elrod (1 May 2026)

Depth Profile Analysis

- 1.73 baseline, confirming a cryosphere dominated by dry, rigid basaltic rock.
- waves diving deeper than 200 km, shift to ratios in the 1.85–2.10 'Volatile-Rich Zone'.
- Reduction in shear strength indicates pore spaces filled with liquid.
- Confirms internal planetary heat likely melts subsurface ice into a global layer of liquid brine

Martian Subsurface Profile: Depth vs. Composition

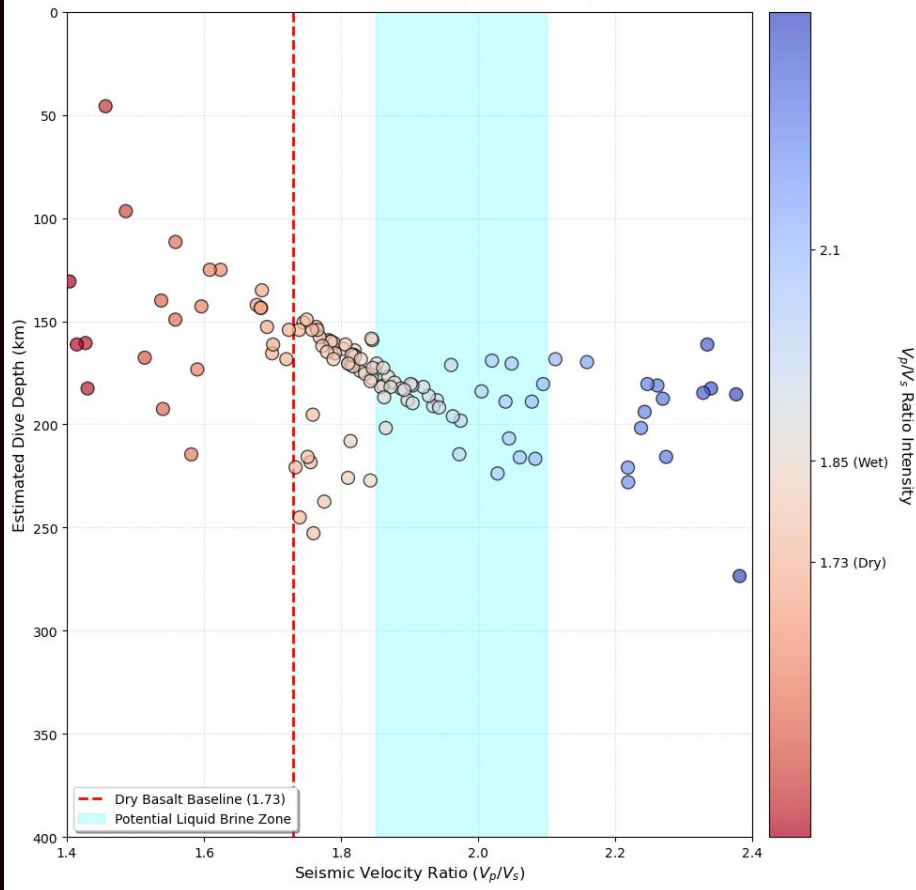


Figure [11]: Depth vs V_p/V_s Ratios by Petra Elrod (1 May 2026)

USING FINDINGS TO MAKE PREDICTIONS

Histogram and Statistics

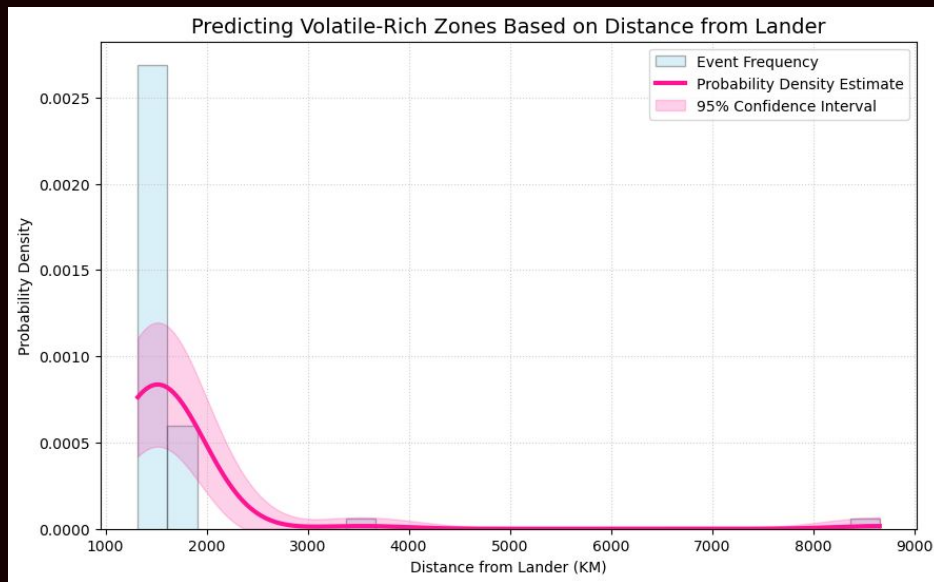


Figure 12: *Predicting Volatile-Rich Zones Based on Distance from Lander* by Addy Peterson(1 May 2026)

Using a histogram

Creates 'bins' to sort by distance.

Probability Density

Trendline shows likelihood of finding a volatile-rich zone at a given distance.

Confidence Interval

Wider CI corresponds to more uncertainty.

Poisson's Ratio

- Measures how much a material thins when stretched or bulges when compressed
- High values indicate fluid-saturated rocks
- Low values indicate stiff rocks
- For our data, Poisson ratios > 0.275 indicate brine saturation in the Martian subsurface
- For comparison, Earth's continental crust has an average Poisson ratio of 0.265 while Earth's oceanic crust has an average Poisson ratio of 0.30 (Hyndman, R. D., 2003)

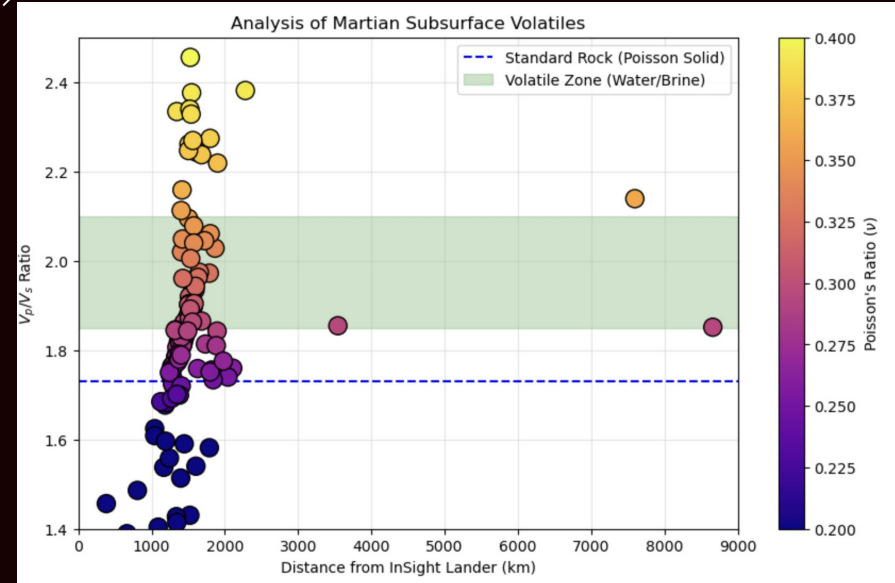


Figure 13: Poisson's ratios compared to V_p/V_s ratios by Shelby Layne (1 May 2026)

CHALLENGES AND WEAKNESSES

Mapping

No triangulation, could not extract azimuth data from files.

Raw Catalog Parsing & Data Extraction

Lack of a pre-formatted data, needed to develop an xml parser.

Seismic Signal-to-Noise Ratio

Environmental noise (wind/thermal); needed to implement a filter to select only certain ratios.

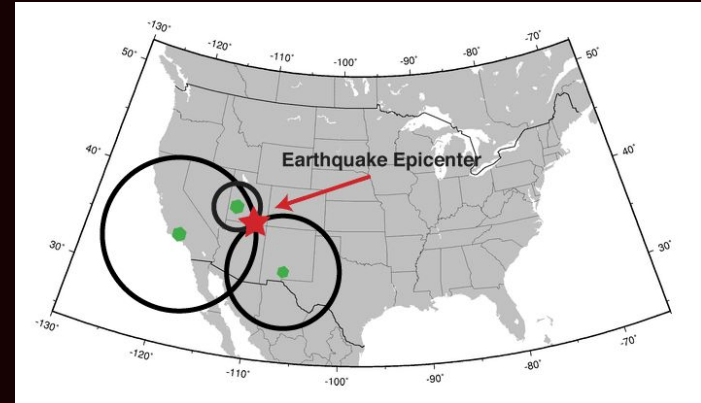


Figure 14: *Triangulation to Locate and Earthquake* by USGS (29 February 2024),
[<https://www.usgs.gov/media/images/triangulation-locate-earthquake>]

SUMMARY

How can variations in V_p/V_s ratios from NASA's Insight data indicate volatile-rich zones in Mars' shallow crust?

- Showing differences between how waves travel through dry, rocky areas compared to volatile-rich zones.
- Revealing localized reductions in shear strength, showing pore spaces filled with liquid.

However...

We cannot directly confirm our findings, but creating predictive models can better inform future missions and data collection.

Future Research

- Confirm our findings of the volatile rich zone with other variables besides V_p/V_s ratios
 - Use density and porosity data from orbital data or Curiosity rover
 - Think about data story 01!
 - Areas with abnormal density decreases and porosity increases were valid locations where gas-hydrates dissociate into water
- Monitor changes in V_p/V_s ratios throughout the Martian year
- If ratios shift with the seasons, could indicate a dynamic hydrological cycle rather than static pockets of water

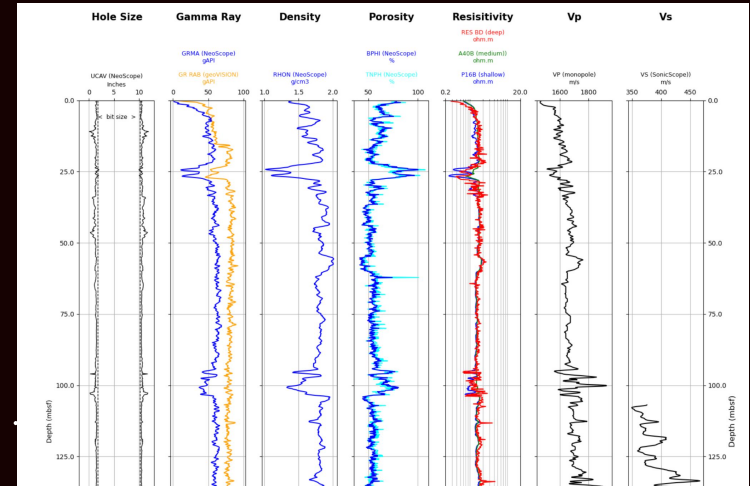


Figure 15: Summary of Logs in Hole U1517A - IODP Expedition 372 by Shelby Layne (8 April 2026)

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Questions